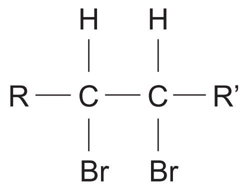
Examiner only

8. (a) Compound **M** is an aliphatic compound, which contains two bromine atoms.



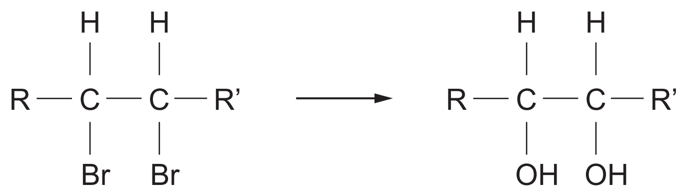
compound **M**

(i) Compound **M** reacts with an excess of alcoholic potassium hydroxide solution to give compound **N**, which has the formula R—C≡C—R'.

Suggest the role of the alcoholic potassium hydroxide in this elimination reaction. [1]

....

(ii) Compound **M** reacts with aqueous sodium hydroxide to give compound **P**.



compound **P**

State the type of reaction mechanism occurring. [1]

....

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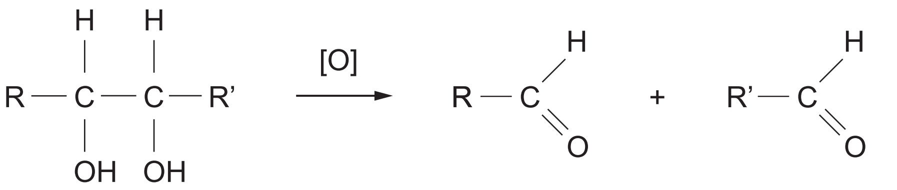
(iii) Compound **P** can undergo bond fission when it is heated with a suitable oxidising agent to form two aldehydes **S** and **T**.



compound **P** compound **S** compound **T**

The mass spectrum of compound **S** showed a molecular ion at m/z 72.

I. The mass spectrum of compound **S** showed that the alkyl group R was branched.

Deduce the structure of compound **S**. Show your reasoning. [2]

II. Compounds **S** and **T** both reacted with 2,4-dinitrophenylhydrazine to give solid derivatives.

State the colour of these derivatives. [1]

....

III. The melting temperatures of the derivatives of compounds **S** and **T** in part II above are shown in the table.

Compound	Melting temperature / °C
S	185–187
T	186–189

Describe how the melting temperature of a 50:50 mixture of the two derivatives differs from those shown in the table. [1]

....

....

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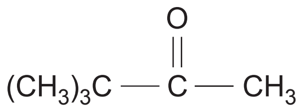
(b) Pinacolone is prepared from a diol containing 6 carbon atoms.



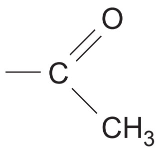
pinacolone

After the reaction, flammable pinacolone is distilled from the reaction mixture. The distillate consists of two layers – an aqueous layer and a less dense layer containing mainly pinacolone.

(i) Describe how you would separate these two layers. [1]

....

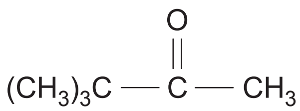
....

(ii) Describe a chemical test, giving reagent(s) and observations, to show that pinacolone contains a  group. [2]

....

....

(iii) The high resolution ^{1}H NMR spectrum of pinacolone consists of two signals.



pinacolone

Complete the table below that describes these two signals. [2]

Protons	Splitting pattern	Relative peak area
CH_3CO		
$(CH_3)_3C$		

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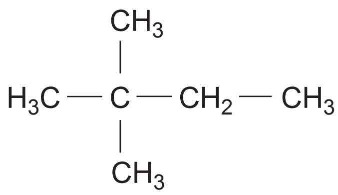
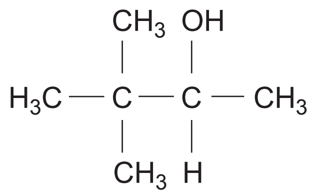
(iv) Under suitable conditions pinacolone was reduced, giving a mixture of 2,2-dimethylbutane and 3,3-dimethylbutan-2-ol together with some unreacted pinacolone.

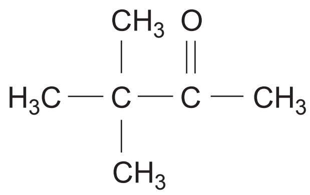
These three compounds were separated by gas chromatography.



2,2-dimethylbutane



3,3-dimethylbutan-2-ol



pinacolone

I. Mary said that the boiling temperature of 3,3-dimethylbutan-2-ol would be the highest of these three compounds and that this could be used to identify it.

Explain why she is correct. [2]

....

....

....

II. The ^{13}C NMR spectra of the remaining two compounds were taken.

Suggest how the ^{13}C NMR spectra of these two compounds would be similar and how they would be different. [2]

Similar ....

....

Different ....

....

15

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